Free-Energy Fluctuations Across the Phase Diagram of Mean-Field Spin Glasses

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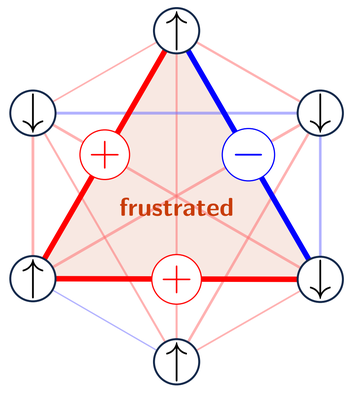
Setup: the Sherrington–Kirkpatrick model

For $\sigma \in \{-1,+1\}^N$ the SK Hamiltonian and free energy are

$$H_N^{\mathrm{SK}}(\sigma) = \frac{\beta}{\sqrt{N}} \sum_{i<j} g_{ij}\,\sigma_i\sigma_j + hNm(\sigma), \quad F_N^{\mathrm{SK}} = \frac{1}{N}\log \sum_{\sigma} e^{H_N^{\mathrm{SK}}(\sigma)}, \quad m(\sigma) = \frac{1}{N}\sum_{i\le N}\sigma_i,$$

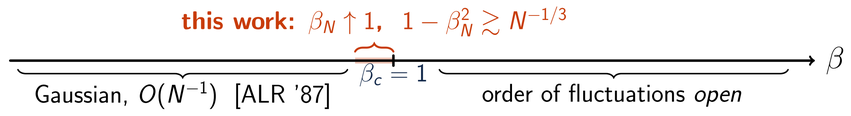
with g_{ij} i.i.d. standard Gaussians.

- **First order — known.** The limit $F^{\mathrm{SK}}(\beta,h) = \lim_N \mathbb{E}F_N^{\mathrm{SK}}$ is given by the **Parisi formula** (Talagrand '06, Panchenko '14): the *law of large numbers*.
- **Next order — this poster.** The *distributional fluctuations* of the free energy. Their scale and shape depend sharply on how the Gibbs measure organizes and, unlike the limit itself, are still only partly understood.



Couplings g_{ij} on every pair: $+$ favors alignment, $-$ opposition; thickness = strength. The shaded triangle $(+,+,-)$ has no satisfying spin assignment — *frustration*. This is what makes the landscape rugged: exponentially many near-optimal states, separated by high barriers.

A. Theorem [DK26]: near the critical temperature



Assume zero-field SK ($h=0$), with β_N in the window $\inf_N N^{1/3}(1-\beta_N^2) > 0$. With $\nu(\beta) = -\frac{1}{2}\log(1-\beta^2) - \frac{\beta^2}{2}$, we have

$$\left|\operatorname{Var}\left(NF_N^{\mathrm{SK}}(\beta_N)\right) - \nu(\beta_N)\right| \;\le\; L\left(N^{1/3}(1-\beta_N^2)\right)^{-3/2},$$

$$N\frac{F_N^{\mathrm{SK}}(\beta_N) - \mathbb{E}F_N^{\mathrm{SK}}(\beta_N)}{\sqrt{\nu(\beta_N)}} \;\Longrightarrow\; \mathcal{N}(0,1) \qquad \text{(with an explicit rate).}$$

Corollary. For $\beta_N = 1 - cN^{-1/3}$:

$$\operatorname{Var}\left(NF_N^{\mathrm{SK}}\right) = \tfrac{1}{6}\log N + O(1) \quad \text{with a Gaussian CLT,}$$

the scale predicted by Aspelmeier '08. Du–Huang '26 reach $\beta = 1$ itself, their variance computation running on the cavity estimates developed here.

A. How it is proved

Everything reduces to one quantity: the *overlap* $R(\sigma,\tau) = \frac{1}{N}\sum_i \sigma_i\tau_i$ of two replicas.

- **Variance representation** via coupled Gaussian interpolation (Chatterjee '09): $\operatorname{Var}\left(NF_N^{\mathrm{SK}}\right) = \frac{\beta^2}{2}\int_0^1 \left(N\,\mathbb{E}\langle R(\sigma,\tau)^2\rangle_t - 1\right) dt$ — the fluctuations *are* an overlap statistic.
- **Cavity method**: decouple the N-th spin and Taylor-expand. Core estimate: $N\,\mathbb{E}\langle R(\sigma,\tau)^2\rangle_t \approx (1-\beta_N^2 t)^{-1}$, with explicit errors. Integrating in t gives $\nu(\beta_N)$, with the error bound stated above.
- **Stein's method**: converts this overlap concentration into a central limit theorem *with a rate*, valid all the way into the critical window.

B. The model: adding a ferromagnetic drive

Add a Curie–Weiss term to the zero-field SK Hamiltonian — the **SK model with ferromagnetic interaction (SKFI)**:

$$H_N(\sigma) = H_N^{\mathrm{SK}}(\sigma)\big|_{h=0} + \underbrace{\frac{\gamma N}{2}\,m(\sigma)^2}_{\text{one stored pattern}} \;=\; \underbrace{\sum_{i<j}\left(\frac{\beta}{\sqrt{N}}\,g_{ij} + \frac{\gamma}{N}\right)\sigma_i\sigma_j}_{\text{couplings i.i.d. } \mathcal{N}(\gamma/N,\,\beta^2/N)} + \frac{\gamma}{2}.$$

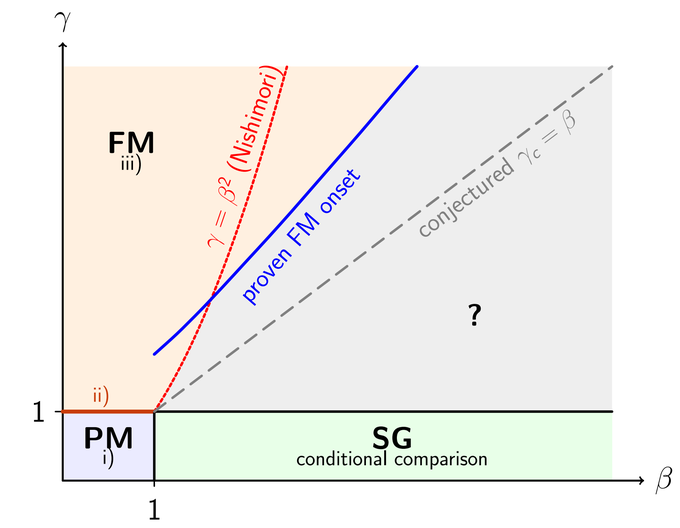
Two readings. *Right:* the couplings are merely *non-centered* — this is Sherrington–Kirkpatrick's original model, γ the ferromagnetic mean, β the disorder strength. *Left:* the drive is *one Hopfield memory*, so the ferromagnetic phase is memory *retrieval*. Disorder and drive compete, producing three phases.

With $F_N = \frac{1}{N}\log\sum_\sigma e^{H_N(\sigma)}$, the limit (Chen '14) and its maximizers are

$$\lim_{N\to\infty} \mathbb{E}F_N(\beta,\gamma) = \max_{\mu\in[-1,1]}\left\{F^{\mathrm{SK}}(\beta,\gamma\mu) - \tfrac{\gamma\mu^2}{2}\right\}, \qquad \Omega(\beta,\gamma) := \arg\max.$$

Paramagnetic (PM): $\beta < 1$ and $\gamma < 1$; then $\Omega = \{0\}$
Ferromagnetic (FM): $\Omega = \{\pm\mu\}$ for some $\mu > 0$
Spin glass (SG): $\beta > 1$ and $\Omega = \{0\}$

B. Phase diagram: where the ferromagnet begins



What is proven about the onset $\gamma_c(\beta)$. It equals 1 for $\beta \le 1$. For $\beta > 1$: $\Omega = \{0\}$ is proven only for $\gamma \le 1$; FM order holds above explicit bounds of *linear* order (slope $\approx$ 1.5–1.6) and above $\gamma = \beta^2$ (Nishimori), pinning the proven region to $(1,1)$. In between: open; conjectured $\gamma_c(\beta) = \beta$.

B. Theorem [DK26+]: a fluctuation trichotomy

Two regimes and the critical line between them — three limit laws.

i) Paramagnetic (fixed $\beta < 1$; $0 \le \gamma_N < 1$ with $\sqrt{N}(1-\gamma_N) \to \infty$): scale N^{-1}, Gaussian.

$$N\left(F_N - \log 2 - \tfrac{\beta^2}{4}\right) + \tfrac{1}{2}\log(1-\gamma_N) \;\Longrightarrow\; \mathcal{N}\left(\tfrac{1}{4}\log(1-\beta^2),\ \nu(\beta)\right), \qquad \text{the same } \nu(\beta) \text{ as in } \textbf{A}.$$

The drive survives only as the deterministic shift $\frac{1}{2}\log(1-\gamma_N)$. Proved for fixed $\gamma < 1$ by Banerjee '20; our short new proof reaches the critical window $\gamma_N \uparrow 1$.

ii) Critical line (fixed $\beta < 1$, and $\gamma_N = 1 + \theta N^{-1/2}$ for $\theta \in \mathbb{R}$): scale N^{-1}, *non-Gaussian*.

$$NF_N - N\log 2 - \tfrac{N-1}{4}\beta^2 - \tfrac{1}{4}\log N \;\Longrightarrow\; \zeta_1 + \log\int_{\mathbb{R}} \exp\left(\left(\zeta_2 + \tfrac{\theta}{2}\right)y^2 - \left(\tfrac{1}{12} + \tfrac{\beta^2}{4(1-\beta^2)}\right)y^4\right)\tfrac{dy}{\sqrt{2\pi}},$$

where $\zeta_1 \sim \mathcal{N}\left(-\frac{\nu(\beta)}{2},\,\nu(\beta)\right)$ and $\zeta_2 \sim \mathcal{N}\left(0,\,\frac{\beta^2}{2(1-\beta^2)}\right)$ are independent.

iii) Ferromagnetic ($\Omega = \{\pm\mu\}$): scale $N^{-1/2}$, Gaussian.

$$\sqrt{N}\left(F_N - \mathbb{E}F_N\right) \;\Longrightarrow\; \mathcal{N}\left(0, \nu_{\mathrm{SK}}(\beta,\gamma\mu)\right),$$

where $\nu_{\mathrm{SK}}(\beta,h) > 0$ is the variance functional of the SK model in external field h (Chen–Dey–Panchenko '17), defined through the Parisi measure. What is actually proved is the stronger statement

$$N \cdot \operatorname{Var}\left(F_N(\beta,\gamma) - F_N^{\mathrm{SK}}(\beta,\gamma\mu)\right) \;\longrightarrow\; 0$$

(the proof works for mixed even p-spin interactions). *Why:* $m(\sigma)$ concentrates exponentially on $\{\pm\mu\}$, so the quadratic drive linearizes — at fluctuation scale SKFI *is* the SK model in the effective field $\gamma\mu$. Inputs: disorder chaos (Chen '13) and magnetization concentration (Chen '14).

B. Spin-glass regime: a conditional comparison

In the remaining regime ($\beta > 1$, $\gamma < 1$) the fluctuations of the zero-field SK model itself are unknown, so instead of computing them we compare the two free energies:

$$F_N(\beta,\gamma) \;\ge\; F_N^{\mathrm{SK}}(\beta,0) \quad \text{a.s.}, \qquad \mathbb{P}\Big(N\big(F_N(\beta,\gamma) - F_N^{\mathrm{SK}}(\beta,0)\big) \ge t\Big) \;\le\; \frac{e^{-t}}{\sqrt{1-\gamma}}.$$

The difference is therefore $O_P(1/N)$ — negligible on any diverging scale. So if $\operatorname{Var}\left(NF_N^{\mathrm{SK}}(\beta,0)\right) \to \infty$ (expected, unproven), then for every fixed $\gamma < 1$ the SKFI free energy obeys *whatever* limit law the zero-field SK model obeys, with the same centering and normalization.

Ising vs. spherical (Baik–Lee '17)

Regime	Spherical SKFI	Ising SKFI (this work)
Paramagnetic	Gaussian, $O(N^{-1})$	Gaussian, $O(N^{-1})$
PM–FM line	Gaussian–TW crossover	non-Gaussian, $O(N^{-1})$
Ferromagnetic	Gaussian, $O(N^{-1/2})$	Gaussian, $O(N^{-1/2})$
Spin glass	TW_1, $O(N^{-2/3})$	= zero-field SK (cond.)
FM–SG boundary	$\gamma = \beta$ (BBP), rigorous	linear order proved

On the sphere, the partition function has an exact random-matrix representation; on the hypercube there is none, which is why the last two rows differ.

Open problems

- FM–SG boundary: with $\varphi_\beta(h) := F^{\mathrm{SK}}(\beta,h) - F^{\mathrm{SK}}(\beta,0)$, $\gamma_c(\beta) = \beta$ is equivalent to $\varphi_\beta(h) \le h^2/(2\beta)$ with equality in the limit $h \downarrow 0$ (Toulouse marginality) — rigorous on the sphere (BBP), open on the hypercube.
- $\beta > 1$: physicists predict $\operatorname{Var}\left(NF_N^{\mathrm{SK}}\right) \sim N^{1/3}$ (Parisi–Rizzo '09), i.e. fluctuations of order $N^{1/6}$. Rigorously, even *divergence* of the variance is open; the best bound is $O(N/\log N)$ (Chatterjee '09).
- Towards the previous item: monotonicity of the coupled overlap in β would give $\operatorname{Var}\left(NF_N^{\mathrm{SK}}\right) \ge \frac{1}{6}\log N - O(1)$ for *all* $\beta \ge 1$.

[DK26] P. S. Dey, T. Kang. *Fluctuations for the SK spin glass model near the critical temperature*, arXiv:2603.05636 (preprint). [DK26+] P. S. Dey, T. Kang. *Fluctuations of the free energy of the Sherrington–Kirkpatrick model with ferromagnetic interaction*, in preparation. Further background: Aizenman–Lebowitz–Ruelle '87; Chatterjee '09; Chen '13, '14; Chen–Dey–Panchenko '17; Baik–Lee '17; Banerjee '20; Dey–Wu '23; Schertzer '26; Du–Huang '26.

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